

Malanka Tharula Wickramasinghe

Bachelor of Computing Honours in Computer Science | University of Sri Jayewardenepura
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EDUCATION

University of Sri Jayewardenepura, Sri Lanka 2023 – 2027 (Expected)

Bachelor of Computing Honours in Computer Science
| Faculty of Computing

Ananda College, Colombo, Sri Lanka Jan 2007 – Dec 2020

G.C.E. Advanced Level — Physical Sciences (Combined Maths, Physics, Chemistry) | Z-Score: 1.5554 | Results: BBB

G.C.E. Ordinary Level — 8 A's, 1 C (incl. English Literature, Tamil, ICT)

TECHNICAL SKILLS

Languages	Python, PHP, Java, C, x86 Assembly, JavaScript, HTML/CSS, SQL
Frameworks & Tools	FastAPI, SQLAlchemy, Alembic, n8n, Docker, Docker Compose, Apache Airflow, Pytest
Databases	PostgreSQL, MySQL, SQLite
AI / ML	Groq API (Llama), Gemini API, LangChain, LangGraph, ChromaDB, Ollama, RAG, Deep Learning (CNN), AI agent building
Dev Tools	Git, GitHub, VS Code, XAMPP, Postman, Figma
Cloud / Deploy	HuggingFace Spaces, GitHub Pages, Google Cloud (OAuth2)
Other	JWT Auth, RBAC, REST API design, ETL pipelines, MVC architecture, Agile/team dev

PROJECTS

DocuAgent — AI Document Extraction Web App

FastAPI · Groq API (Llama 4 Scout) · pdf2image · ReportLab · Docker · GitHub Pages · HuggingFace Spaces

- Full-stack app: upload PDF/PNG/JPG → AI extracts summary, key fields, and anomalies
- Exports structured results to Excel and PDF report
- Backend deployed on HuggingFace Spaces (Docker); frontend on GitHub Pages

HR Onboarding Agent — AI Automation Pipeline

n8n · Groq API (Llama 3.3 70B) · Gmail OAuth2 · Google Sheets OAuth2 · FastAPI · Docker

- Single form submission triggers full onboarding workflow via n8n webhook
- AI generates personalised welcome email + task checklist; auto-sends via Gmail API
- Logs all records to Google Sheets; notifies HR automatically

EmployeeAPI — Production-Grade REST API

FastAPI · PostgreSQL · SQLAlchemy · Alembic · JWT · bcrypt · Docker Compose · Pytest

- Full CRUD REST API with JWT Bearer token auth and role-based access control (admin/user)
- Alembic database migrations; self-referential manager relationship in data model
- 10 passing Pytest tests; two-container Docker Compose (API + PostgreSQL 16)

FinDataPipeline — Financial Data ETL Pipeline

Apache Airflow 2.9 · PostgreSQL · FastAPI · SQLAlchemy · Chart.js · Docker Compose

- Automated pipeline: Alpha Vantage API → extract → transform (moving averages, % change) → PostgreSQL
- Scheduled by Airflow (weekday runs after US market close); exposes data via FastAPI
- Live dashboard with Chart.js charts; tracks AAPL, MSFT, GOOGL, JPM, TSLA

CodeRAG — Local Codebase RAG System

FastAPI · LangChain · ChromaDB · Ollama (Llama 3.2 3B · nomic-embed-text) · HTML/CSS/JS

- Fully local RAG system — no cloud, no API keys; GitHub URL → clone → code-aware chunking → ChromaDB → Q&A
- Retrieves top-5 relevant chunks; Llama 3.2 3B (Ollama) answers with full code context
- Runs entirely on local GPU (GTX 1650 4GB); fully private — no data leaves the machine

MultiAgentDev — AI Multi-Agent Developer System

LangGraph · Groq API (Llama 3.3 70B) · FastAPI · LangChain · HuggingFace Spaces · HTML/CSS/JS

- LangGraph StateGraph orchestrator classifies tasks and routes to specialist agents: Code, Debug, Doc, Test
- Each agent reads shared AgentState (TypedDict) and writes results back; conditional edges prevent cross-contamination
- Live demo deployed on HuggingFace Spaces; frontend with syntax highlighting, markdown render, dark mode

ASL-to-Text — Real-Time Sign Language Recognition

Python · Deep Learning (CNN) · Computer Vision | Group Project

- Deep learning model for real-time American Sign Language recognition trained on 200k+ images
- Achieved high accuracy on test set; built for accessibility applications

Library Management System (LMS)

PHP (MVC) · MySQL · Bootstrap · AJAX · RBAC | Group Project — Backend Developer

- Full-featured LMS with role-based access control (Admin, Staff, Member)
- Features: book reservation, borrow history, approval workflows, email notifications

Custom Operating System

C · x86 Assembly · QEMU | Group Project

- Custom bootloader + kernel, basic GUI, keyboard/mouse input, simple task system — no existing OS

RESEARCH & PUBLICATIONS

Published Paper — Scientific Journal of Economics, Management and Sustainability (JEMS), 2024

"A dual-phase financial strategy for sustainable customer loyalty: Integrating fixed deposits and voucher systems in Sri Lankan supermarket retail"

ACHIEVEMENTS & AWARDS

- **Team CodeDuoPlus One:** IEEEExtreme 18.0 (24-hour global coding competition, 8,000+ teams) — Global Rank: 605 | Country Rank: 41 | University Rank: 1st
- Only USJ representative invited to IFS Champions Day 2025 at Orion City — exclusive event for Sri Lanka's top-ranked IEEEExtreme coders
- Co-founder & first Chief Editor, Association of Computing Students – USJ; led CodeFuse hackathon, TechXplore industry event, EDUACS seminar series, ACS eMagazine

LEADERSHIP & VOLUNTEERING

Chief Editor — Association of Computing Students, USJ Apr 2024 – Jun 2025

Shaped brand identity; led publications, editorial team, and all major events

Design & Marketing Head — J'PURAXTREME 2.0, IEEE CS Student Branch USJ 2024

Led all design and marketing efforts for the university's flagship tech event

Member — IEEE Student Branch, University of Sri Jayewardenepura Oct 2023 – Present

CERTIFICATIONS

- Introduction to Software Engineering — Udemy (Feb 2024)
- Java Programming Masterclass: Beginner to Master — Udemy (Feb 2024)
- Introduction to Database Management Systems — Udemy (Feb 2024)
- Master Course of Rapid Application Development — Udemy (Feb 2024)
- Introduction to Programming Using Java — Sololearn (Jan 2024)

LANGUAGES

English (Professional) | Sinhala (Native)